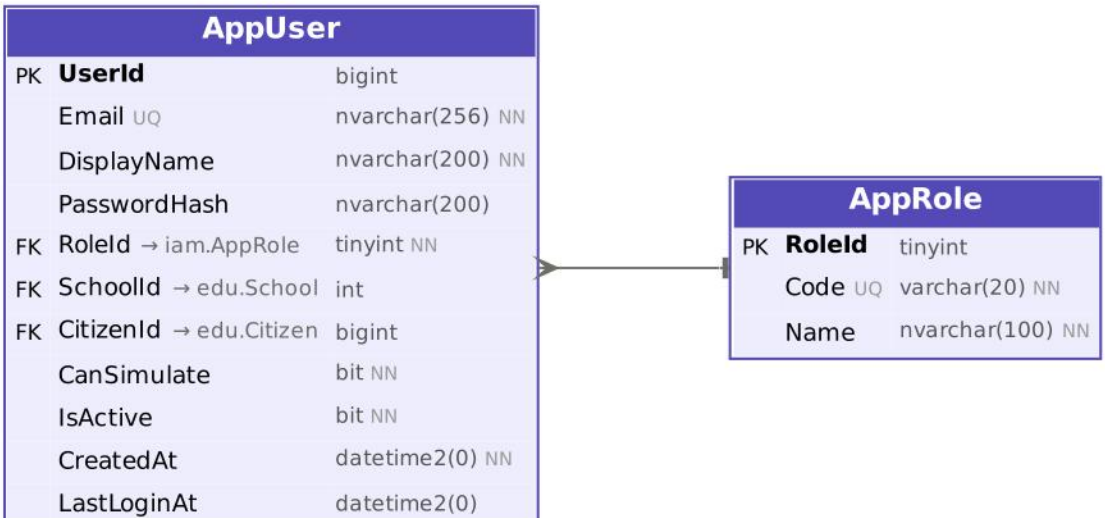
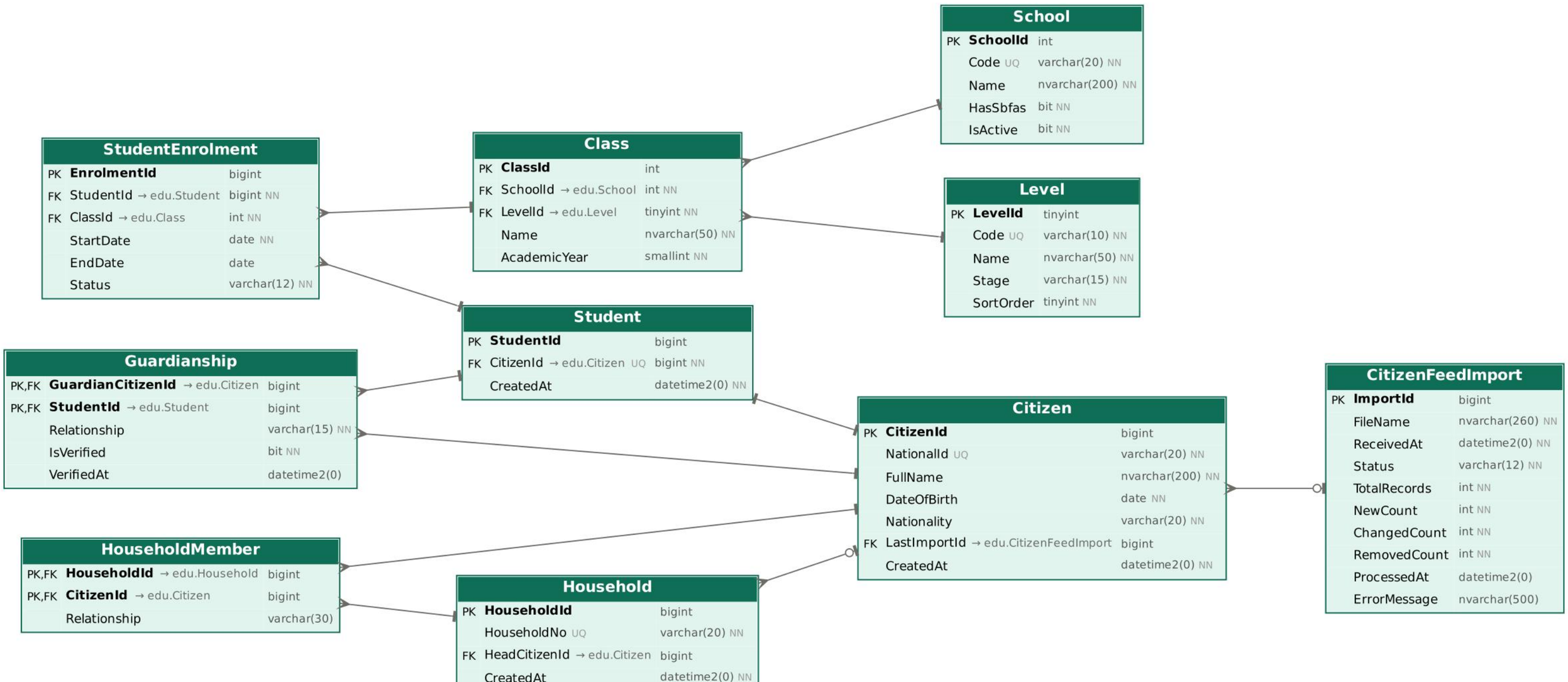
iam - Người dùng và vai trò



edu - Trường, học sinh, hộ gia đình



fin - Quỹ, tài khoản, top-up, học phí, CPF

Fund

PK	**FundId**	smallint
	Code UQ	varchar(30) NN
	Name	nvarchar(100) NN
	Balance	decimal(18,2) NN
	RowVer	rowversion NN

TopUpBatch

PK	**BatchId**	bigint
FK	RuleId → fin.TopUpRule	int
FK	ScheduleId → fin.TopUpSchedule	int
FK	FundId → fin.Fund	smallint NN
	BatchType	varchar(5) NN
	Status	varchar(16) NN
	TotalCount	int NN
	SuccessCount	int NN
	FailedCount	int NN
	TotalAmount	decimal(18,2) NN
	IdempotencyKey UQ	varchar(100) NN
FK	RequestedByUserId → iam.AppUser	bigint NN
FK	ApprovedByUserId → iam.AppUser	bigint
	ApprovedAt	datetime2(0)
	CreatedAt	datetime2(0) NN
	StartedAt	datetime2(0)
	CompletedAt	datetime2(0)

TopUpSchedule

PK	**ScheduleId**	int
FK	RuleId → fin.TopUpRule	int NN
	Frequency	varchar(10) NN
	NextRunAt	datetime2(0) NN
	LastRunAt	datetime2(0)
	IsActive	bit NN

TopUpRule

PK	**RuleId**	int
	Name	nvarchar(150) NN
	TriggerType	varchar(16) NN
	Amount	decimal(18,2) NN
	MinAge	tinyint
	MaxAge	tinyint
	RequireEnrolled	bit NN
	UseBalanceCondition	bit NN
	BalanceBelow	decimal(18,2)
	BalanceCap	decimal(18,2)
	RequireApproval	bit NN
	EffectiveFrom	date NN
	EffectiveTo	date
	IsActive	bit NN
FK	CreatedByUserId → iam.AppUser	bigint NN
	CreatedAt	datetime2(0) NN

TopUpItem

PK	**ItemId**	bigint
FK	BatchId → fin.TopUpBatch	bigint NN
FK	AccountId → fin.EducationAccount	bigint NN
	Amount	decimal(18,2) NN
	Status	varchar(10) NN
	ErrorMessage	nvarchar(400)
FK	TransactionId → fin.AccountTransaction	bigint

AccountTransaction

PK	**TransactionId**	bigint
FK	AccountId → fin.EducationAccount	bigint NN
	TxnType	varchar(20) NN
	Amount	decimal(18,2) NN
	BalanceAfter	decimal(18,2) NN
	ReferenceType	varchar(30)
	ReferenceId	bigint
	Description	nvarchar(300)
	IdempotencyKey	varchar(100)
FK	CreatedByUserId → iam.AppUser	bigint
	CreatedAt	datetime2(0) NN

EducationAccount

PK	**AccountId**	bigint
FK	CitizenId → edu.Citizen UQ	bigint NN
	Status	varchar(10) NN
	Balance	decimal(18,2) NN
	OpenedAt	datetime2(0) NN
	ClosedAt	datetime2(0)
	ClosureReason	varchar(20)
	RowVer	rowversion NN

CpfTransfer

PK	**TransferId**	bigint
FK	AccountId → fin.EducationAccount	bigint NN
FK	CpfAccountId → fin.CpfAccount	bigint NN
	Amount	decimal(18,2) NN
	Status	varchar(10) NN
	ExternalRef	varchar(100)
	ErrorMessage	nvarchar(400)
FK	TransactionId → fin.AccountTransaction	bigint
	RequestedAt	datetime2(0) NN
	CompletedAt	datetime2(0)

CpfAccount

PK	**CpfAccountId**	bigint
FK	CitizenId → edu.Citizen UQ	bigint NN
	CpfAccountNo UQ	varchar(20) NN
	CreatedAt	datetime2(0) NN

Payment

PK	**PaymentId**	bigint
FK	ChargeId → fin.StudentCharge	bigint NN
FK	PayerUserId → iam.AppUser	bigint NN
	Method	varchar(12) NN
	Channel	varchar(10)
	Amount	decimal(18,2) NN
	Status	varchar(10) NN
	GatewayRef	varchar(100)
	IdempotencyKey	varchar(100)
FK	TransactionId → fin.AccountTransaction	bigint
	CreatedAt	datetime2(0) NN
	CompletedAt	datetime2(0)

FeeTypePayableRule

PK	**PayableRuleId**	int
FK	FeeTypeId → fin.FeeType	smallint NN
	PayableByEduAccount	bit NN
	EffectiveFrom	date NN
	EffectiveTo	date
FK	CreatedByUserId → iam.AppUser	bigint NN

FeeType

PK	**FeeTypeId**	smallint
	Code UQ	varchar(30) NN
	Name	nvarchar(100) NN
	IsActive	bit NN

StudentCharge

PK	**ChargeId**	bigint
FK	StudentId → edu.Student	bigint NN
FK	FeeComponentId → fin.FeeComponent	int NN
	AmountDue	decimal(18,2) NN
	SubsidyAmount	decimal(18,2) NN
	AmountPaid	decimal(18,2) NN
	NetPayable	computed
	Outstanding	computed
	Status	varchar(10) NN
	DueDate	date NN
	CreatedAt	datetime2(0) NN

FeeComponent

PK	**FeeComponentId**	int
FK	ClassId → edu.Class	int NN
FK	FeeTypeId → fin.FeeType	smallint NN
	Term	tinyint
	Amount	decimal(18,2) NN
	DueDate	date NN

ChargeSubsidy

PK	**ChargeSubsidyId**	bigint
FK	ChargeId → fin.StudentCharge	bigint NN
FK	ResultId → fas.ChildResult	bigint NN
	Amount	decimal(18,2) NN

fas - Hỗ trợ tài chính (FAS)

RiskAssessment

PK	**RiskId**	bigint
FK	ApplicationId → fas.Application	bigint NN
	RiskLevel	varchar(6) NN
	Probability	decimal(5,4) NN
	SuggestedAmount	decimal(18,2)
	ModelVersion	varchar(30) NN
	CreatedAt	datetime2(0) NN

ApplicationChild

PK	**ApplicationChildId**	bigint
FK	ApplicationId → fas.Application	bigint NN
FK	StudentId → edu.Student	bigint NN

ApplicationFieldValue

PK	**FieldValueId**	bigint
FK	ApplicationId → fas.Application	bigint NN
	FieldKey	varchar(60) NN
	AiSuggestedValue	nvarchar(500)
	FinalValue	nvarchar(500)
	Source	varchar(15) NN
	IsLocked	bit NN
	UpdatedAt	datetime2(0) NN

Application

PK	**ApplicationId**	bigint
FK	HouseholdId → edu.Household	bigint NN
	AcademicYear	smallint NN
	Channel	varchar(6) NN
FK	SubmittedByUserId → iam.AppUser	bigint
FK	EnteredByUserId → iam.AppUser	bigint
FK	IncomeSnapshotId → fas.HouseholdIncomeSnapshot	bigint
	Status	varchar(10) NN
	CreatedAt	datetime2(0) NN
	SubmittedAt	datetime2(0)
	RowVer	rowversion NN

PaperScan

PK	**ScanId**	bigint
FK	ApplicationId → fas.Application	bigint NN
	BlobUri	nvarchar(500) NN
	OcrStatus	varchar(10) NN
FK	DetectedSchemeId → fas.Scheme	tinyint
	OcrConfidence	decimal(5,4)
FK	UploadedByUserId → iam.AppUser	bigint NN
	UploadedAt	datetime2(0) NN

SbfasWorkflowStep

PK	**WorkflowStepId**	int
FK	SchoolId → edu.School	int NN
	StepNo	tinyint NN
	Name	nvarchar(100) NN
FK	ReviewerRoleId → iam.AppRole	tinyint NN

SbfasReviewStep

PK	**ReviewStepId**	bigint
FK	ResultId → fas.ChildResult	bigint NN
	StepNo	tinyint NN
	StepName	nvarchar(100) NN
FK	ReviewerUserId → iam.AppUser	bigint
	Decision	varchar(10) NN
	Comment	nvarchar(500)
	DecidedAt	datetime2(0)

ChildResult

PK	**ResultId**	bigint
FK	ApplicationChildId → fas.ApplicationChild	bigint NN
FK	SchemeId → fas.Scheme	tinyint NN
FK	PolicyVersionId → fas.PolicyVersion	int NN
FK	TierId → fas.BenefitTier	int
	Status	varchar(15) NN
	TotalAmount	decimal(18,2) NN
	Reason	nvarchar(400)
	DecidedAt	datetime2(0)
FK	DecidedByUserId → iam.AppUser	bigint

HouseholdIncomeSnapshot

PK	**SnapshotId**	bigint
FK	HouseholdId → edu.Household	bigint NN
FK	HouseholdBandId → fas.IncomeBand	smallint NN
FK	PerCapitaBandId → fas.IncomeBand	smallint NN
	MemberCount	tinyint NN
	Source	varchar(20) NN
	FetchedAt	datetime2(0) NN

PolicyVersion

PK	**PolicyVersionId**	int
FK	SchemeId → fas.Scheme	tinyint NN
FK	SchoolId → edu.School	int
	VersionNo	int NN
	RequiredNationality	varchar(20)
	RequireEnrolled	bit NN
	EffectiveFrom	date NN
	EffectiveTo	date
	Status	varchar(10) NN
FK	CreatedByUserId → iam.AppUser	bigint NN
	CreatedAt	datetime2(0) NN

Scheme

PK	**SchemeId**	tinyint
	Code UQ	varchar(20) NN
	Name	nvarchar(150) NN
	Scope	varchar(10) NN

IncomeBand

PK	**IncomeBandId**	smallint
	BandType	varchar(10) NN
	Code	varchar(10) NN
	Name	nvarchar(60) NN
	MinAmount	decimal(18,2)
	MaxAmount	decimal(18,2)
	SortOrder	tinyint NN

TierBenefit

PK,FK	**TierId** → fas.BenefitTier	int
PK,FK	**FeeTypeId** → fin.FeeType	smallint
	Amount	decimal(18,2) NN
	Frequency	varchar(5) NN

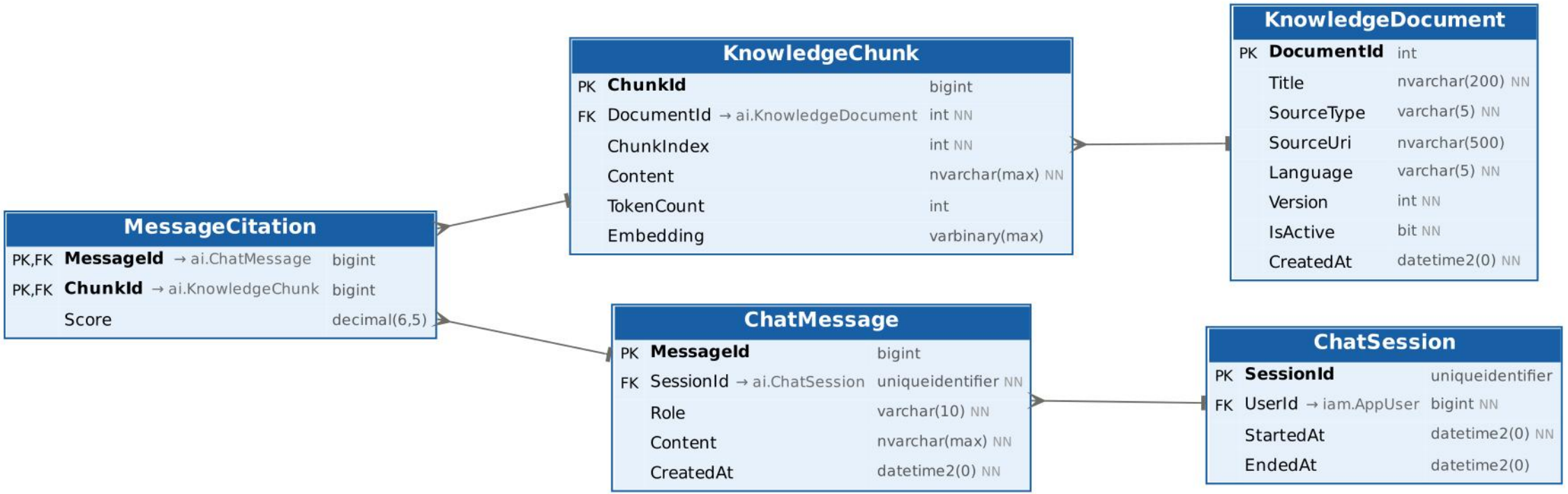
BenefitTier

PK	**TierId**	int
FK	PolicyVersionId → fas.PolicyVersion	int NN
	TierNo	tinyint NN
	Name	nvarchar(50) NN
FK	MaxPerCapitaBandId → fas.IncomeBand	smallint
FK	MaxHouseholdBandId → fas.IncomeBand	smallint

Budget

PK	**BudgetId**	int
FK	SchemeId → fas.Scheme	tinyint NN
FK	SchoolId → edu.School	int
	AcademicYear	smallint NN
	TotalBudget	decimal(18,2) NN
	UsedAmount	decimal(18,2) NN
FK	CreatedByUserId → iam.AppUser	bigint NN

ai - Chatbot RAG



aud - Audit và simulate

